Northeastern University
College *of* Professional Studies

ALY6015 - Intermediate Analytics

Analysis of Telecom Customer Churn Prediction Dataset

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1. Introduction

The main goal of the assignment is to take the theoretical knowledge learned from this course and apply it to real-world data and use data analysis to solve valuable problems. For this final group project, we've selected the Telecom Customer Churn Prediction Dataset. The datasets contain 2 tables in separate csv files.

- **The Customer Churn table**: Contains information on all 7,043 customers and 38 variables from a Telecommunications company in California in Q2 2022. Each record represents one customer, and contains details about their demographics, location, tenure, subscription services, status for the quarter (joined, stayed, or churned), and more.
- **The Zip Code Population table**: Contains complimentary information on the estimated populations for the California zip codes in the Customer Churn table.

2. Telecom Dataset

2.1 Telecom_customer_churn dataset:

1. **Customer ID:** A unique ID that identifies each customer.
2. **Gender:** The customer's gender: Male, Female.
3. **Age:** The customer's current age, in years, at the time the fiscal quarter ended (Q2 2022).
4. **Married:** Indicates if the customer is married: Yes, No.
5. **Number of Dependents:** Indicates the number of dependents that live with the customer (dependents could be children, parents, grandparents, etc.).
6. **City:** The city of the customer's primary residence in California.
7. **Zip Code:** The zip code of the customer's primary residence.
8. **Latitude:** The latitude of the customer's primary residence.

9. **Longitude:** The longitude of the customer's primary residence.
10. **Number of Referrals:** Indicates the number of times the customer has referred a friend or family member to this company to date.
11. **Tenure in Months:** Indicates the total amount of months that the customer has been with the company by the end of the quarter specified above.
12. **Offer:** Identifies the last marketing offer that the customer accepted: None, Offer A, Offer B, Offer C, Offer D, Offer E.
13. **Phone Service:** Indicates if the customer subscribes to home phone service with the company: Yes, No.
14. **Avg Monthly Long-Distance Charges:** Indicates the customer's average long-distance charges, calculated to the end of the quarter specified above (if the customer is not subscribed to home phone service, this will be 0).
15. **Multiple Lines:** Indicates if the customer subscribes to multiple telephone lines with the company: Yes, no (if the customer is not subscribed to home phone service, this will be No)
16. **Internet Service:** Indicates if the customer subscribes to Internet service with the company: Yes, No.
17. **Internet Type:** Indicates the customer's type of internet connection: DSL, Fiber Optic, Cable (if the customer is not subscribed to internet service, this will be None).
18. **Avg Monthly GB Download:** Indicates the customer's average download volume in gigabytes, calculated to the end of the quarter specified above (if the customer is not subscribed to internet service, this will be 0).

19. **Online Security:** Indicates if the customer subscribes to an additional online security service provided by the company: Yes, no (if the customer is not subscribed to internet service, this will be No).
20. **Online Backup:** Indicates if the customer subscribes to an additional online backup service provided by the company: Yes, no (if the customer is not subscribed to internet service, this will be No).
21. **Device Protection Plan:** Indicates if the customer subscribes to an additional device protection plan for their Internet equipment provided by the company: Yes, no (if the customer is not subscribed to internet service, this will be No).
22. **Premium Tech Support:** Indicates if the customer subscribes to an additional technical support plan from the company with reduced wait times: Yes, no (if the customer is not subscribed to internet service, this will be No).
23. **Streaming TV:** Indicates if the customer uses their Internet service to stream television programing from a third-party provider at no additional fee: Yes, no (if the customer is not subscribed to internet service, this will be No).
24. **Streaming Movies:** Indicates if the customer uses their Internet service to stream movies from a third-party provider at no additional fee: Yes, no (if the customer is not subscribed to internet service, this will be No).
25. **Streaming Music:** Indicates if the customer uses their Internet service to stream music from a third-party provider at no additional fee: Yes, no (if the customer is not subscribed to internet service, this will be No).
26. **Unlimited Data:** Indicates if the customer has paid an additional monthly fee to have unlimited data downloads/uploads: Yes, no (if the customer is not subscribed to internet service, this will be No).

27. **Contract:** Indicates the customer's current contract type: Month-to-Month, One Year, Two Year.
28. **Paperless Billing:** Indicates if the customer has chosen paperless billing: Yes, No.
29. **Payment Method:** Indicates how the customer pays their bill: Bank Withdrawal, Credit Card, Mailed Check.
30. **Monthly Charge:** Indicates the customer's current total monthly charge for all their services from the company.
31. **Total Charges:** Indicates the customer's total charges, calculated to the end of the quarter specified above.
32. **Total Refunds:** Indicates the customer's total refunds, calculated to the end of the quarter specified above.
33. **Total Extra Data Charges:** Indicates the customer's total charges for extra data downloads above those specified in their plan, by the end of the quarter specified above.
34. **Total Long-Distance Charges:** Indicates the customer's total charges for long distance above those specified in their plan, by the end of the quarter specified above.
35. **Total Revenue:** Indicates the company's total revenue from this customer, calculated to the end of the quarter specified above (Total Charges - Total Refunds + Total Extra Data Charges + Total Long-Distance Charges)
36. **Customer Status:** Indicates the status of the customer at the end of the quarter: Churned, Stayed, or Joined.

37. **Churn Category:** A high-level category for the customer's reason for churning, which is asked when they leave the company: Attitude, Competitor, Dissatisfaction, Other, Price (directly related to Churn Reason).
38. **Churn Reason:** A customer's specific reason for leaving the company, which is asked when they leave the company (directly related to Churn Category).

2.2 Telecom_zipcode_population dataset:

1. **Zip Code:** The zip code of the customer's primary residence.
2. **Population:** A current population estimate for the entire Zip Code area.

3. Research Objectives and Questions

Below are the questions we will try to analyze:

<u>Cheng</u>:

- **Churn Finding:** What are the most important factors that contribute to high retention?
- **Customer Segmentation:** Divide the customer base into different groups and find marketing strategies.

<u>Chandana</u>:

- **Service Usage Analysis:** Analyze service usage patterns within the dataset to understand which services are most popular among different demographic groups and which services are prone to high churn rates.
- **Model Validation:** Which analysis model can accurately predict customer churn rate?

This analysis aims to predict customer churn more accurately and also hopes to provide a deeper analysis of customer behavior and preferences to help telecoms companies develop more

effective customer retention and service strategies. We will apply a variety of analysis techniques in the course, such as the Chi-Square, ANOVA and Logistic Regression, trying to solve the four questions we listed or the potential influencing factors. For this collaborative project with Chandana, I will handle tasks 1 and 2, and she will take care of tasks 3 and 4.

4. Exploratory Data Analysis (by Cheng)

4.1 Initial Analysis

For more efficient analysis, we merged the two datasets, the following is the summary of merged telecom dataset:

```
> summary(merged_telecom)
    Zip.Code     Customer.ID          Gender               Age          Married
 Min.   :90001   Length:7043        Length:7043        Min.   :19.00   Length:7043
 1st Qu.:92101   Class :character   Class :character   1st Qu.:32.00   Class :character
 Median :93518   Mode  :character   Mode  :character   Median :46.00   Mode  :character
 Mean   :93486                                         Mean   :46.51
 3rd Qu.:95329                                         3rd Qu.:60.00
 Max.   :96150                                         Max.   :80.00

 Number.of.Dependents     City              Latitude       Longitude      Number.of.Referrals
 Min.   :0.0000       Length:7043        Min.   :32.56   Min.   :-124.3   Min.   : 0.000
 1st Qu.:0.0000       Class :character   1st Qu.:33.99   1st Qu.:-121.8   1st Qu.: 0.000
 Median :0.0000       Mode  :character   Median :36.21   Median :-119.6   Median : 0.000
 Mean   :0.4687                          Mean   :36.20   Mean   :-119.8   Mean   : 1.952
 3rd Qu.:0.0000                          3rd Qu.:38.16   3rd Qu.:-118.0   3rd Qu.: 3.000
 Max.   :9.0000                          Max.   :41.96   Max.   :-114.2   Max.   :11.000

 Tenure.in.Months    Offer           Phone.Service      Avg.Monthly.Long.Distance.Charges
 Min.   : 1.00    Length:7043        Length:7043        Min.   : 1.01
 1st Qu.: 9.00    Class :character   Class :character   1st Qu.:13.05
 Median :29.00    Mode  :character   Mode  :character   Median :25.69
 Mean   :32.39                                          Mean   :25.42
 3rd Qu.:55.00                                          3rd Qu.:37.68
 Max.   :72.00                                          Max.   :49.99
                                                        NA's   :682
 Multiple.Lines     Internet.Service   Internet.Type      Avg.Monthly.GB.Download Online.Security
 Length:7043        Length:7043        Length:7043        Min.   : 2.00           Length:7043
 Class :character   Class :character   Class :character   1st Qu.:13.00           Class :character
 Mode  :character   Mode  :character   Mode  :character   Median :21.00           Mode  :character
                                                           Mean   :26.19
                                                           3rd Qu.:30.00
                                                           Max.   :85.00
                                                           NA's   :1526
```

```
Online.Backup      Device.Protection.Plan Premium.Tech.Support Streaming.TV
Length:7043        Length:7043            Length:7043          Length:7043
Class :character   Class :character       Class :character     Class :character
Mode  :character   Mode  :character       Mode  :character     Mode  :character


Streaming.Movies   Streaming.Music    Unlimited.Data       Contract         Paperless.Billing
Length:7043        Length:7043        Length:7043        Length:7043        Length:7043
Class :character   Class :character   Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character   Mode  :character   Mode  :character


Payment.Method     Monthly.Charge   Total.Charges    Total.Refunds    Total.Extra.Data.Charges
Length:7043        Min.   :-10.00   Min.   :  18.8   Min.   : 0.000   Min.   :  0.000
Class :character   1st Qu.: 30.40   1st Qu.: 400.1   1st Qu.: 0.000   1st Qu.:  0.000
Mode  :character   Median : 70.05   Median :1394.5   Median : 0.000   Median :  0.000
                   Mean   : 63.60   Mean   :2280.4   Mean   : 1.962   Mean   :  6.861
                   3rd Qu.: 89.75   3rd Qu.:3786.6   3rd Qu.: 0.000   3rd Qu.:  0.000
                   Max.   :118.75   Max.   :8684.8   Max.   :49.790   Max.   :150.000

Total.Long.Distance.Charges Total.Revenue      Customer.Status    Churn.Category
Min.   :   0.00             Min.   :   21.36   Length:7043        Length:7043
1st Qu.:  70.55             1st Qu.:  605.61   Class :character   Class :character
Median : 401.44             Median : 2108.64   Mode  :character   Mode  :character
Mean   : 749.10             Mean   : 3034.38
3rd Qu.:1191.10             3rd Qu.: 4801.15
Max.   :3564.72             Max.   :11979.34

Churn.Reason         Population
Length:7043        Min.   :    11
Class :character   1st Qu.:  2344
Mode  :character   Median : 17554
                   Mean   : 22140
                   3rd Qu.: 36125
                   Max.   :105285
```

⬆️ The new merged_telecomdataset contains 39 columns and 7,043 rows.

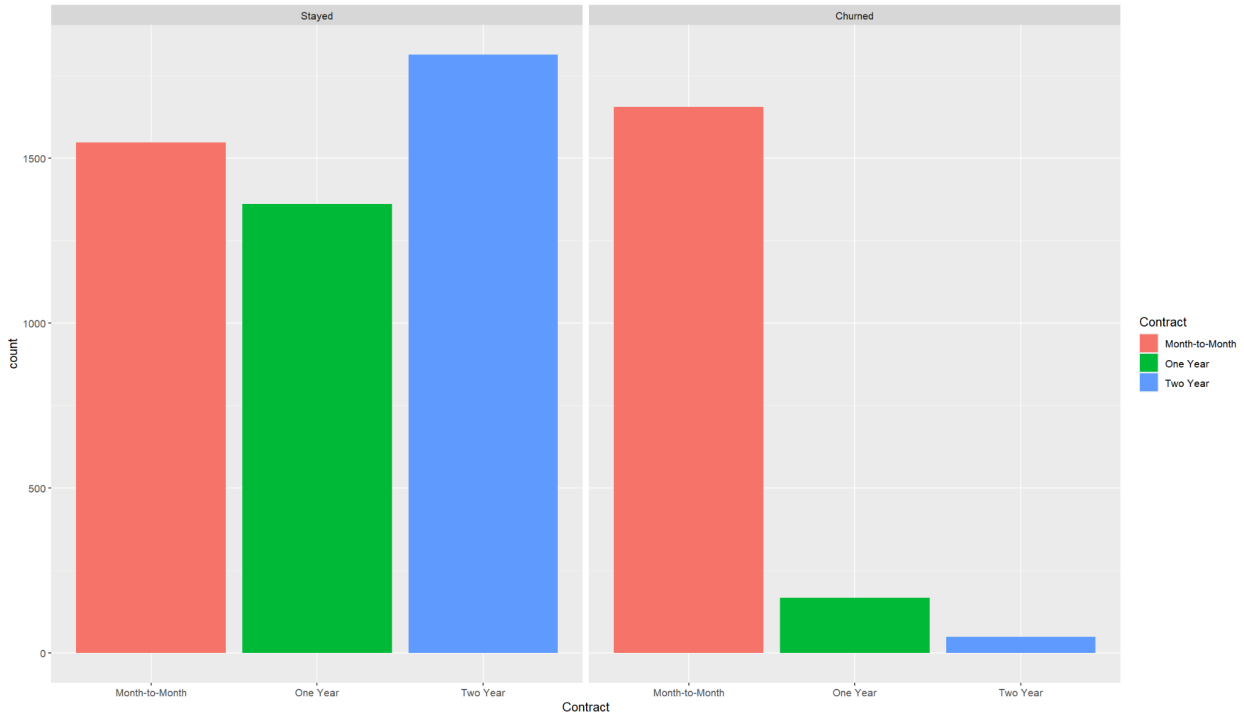
4.2 Missing Values:

```
> missing_summary
                          Zip.Code                          Customer.ID
                                 0                                    0
                            Gender                                  Age
                                 0                                    0
                           Married                 Number.of.Dependents
                                 0                                    0
                              City                             Latitude
                                 0                                    0
                         Longitude                  Number.of.Referrals
                                 0                                    0
                  Tenure.in.Months                                Offer
                                 0                                    0
                     Phone.Service Avg.Monthly.Long.Distance.Charges
                                 0                                  682
                    Multiple.Lines                     Internet.Service
                                 0                                    0
                     Internet.Type              Avg.Monthly.GB.Download
                                 0                                 1526
                   Online.Security                       Online.Backup
                                 0                                    0
             Device.Protection.Plan                 Premium.Tech.Support
                                 0                                    0
                       Streaming.TV                     Streaming.Movies
                                 0                                    0
                   Streaming.Music                       Unlimited.Data
                                 0                                    0
                          Contract                    Paperless.Billing
                                 0                                    0
                    Payment.Method                       Monthly.Charge
                                 0                                    0
                     Total.Charges                        Total.Refunds
                                 0                                    0
           Total.Extra.Data.Charges         Total.Long.Distance.Charges
                                 0                                    0
                     Total.Revenue                      Customer.Status
                                 0                                    0
                    Churn.Category                         Churn.Reason
                                 0                                    0
                        Population
                                 0
```

⬆️ Avg.Monthly. Long.Distance.Charges has 682 missing values, and

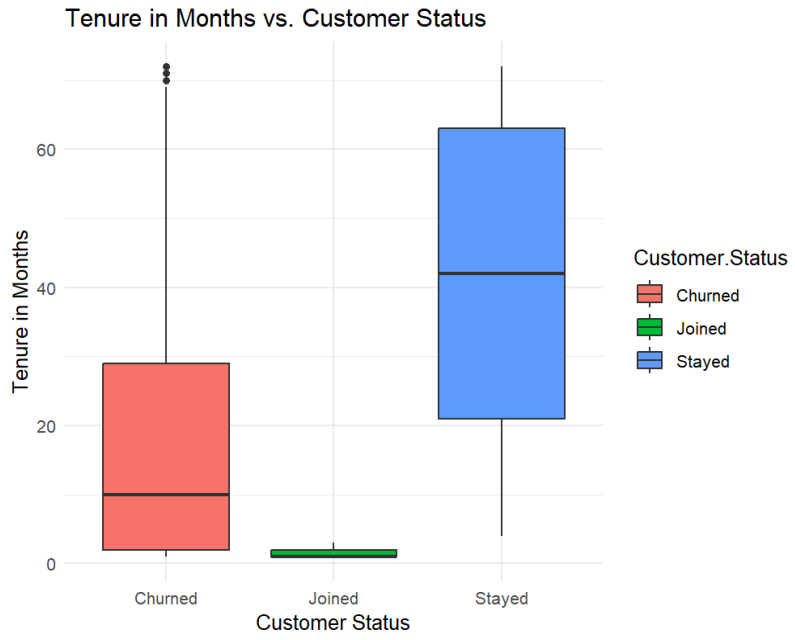
Avg.Monthly.GB.Download has 1,526 missing values.

4.3 Bar plot for contract vs. customer status:



⬆️ As can be seen from the chart above, the majority of churned customers were on month-to-month contracts, while fewer customers on one- and two-year contracts churned, suggesting that shorter month-to-month contracts are associated with higher churn rates, which may be due to lower exit barriers for short term customers.

4.4 Box plot for tenure in months vs. customer status:



⬆️ As can be seen from the chart above:

Churned Customers:

- Median tenure: The line within the red box indicates that half of the churned customers who have tenure of about 10 months or less.
- IQR: The red box shows the middle 50% of churned customers with tenure of approximately 3 to 30 months.
- Outliers: The black dots above the boxes represent outliers, indicating that some churned customers had significantly longer tenures.

Joined Customers:

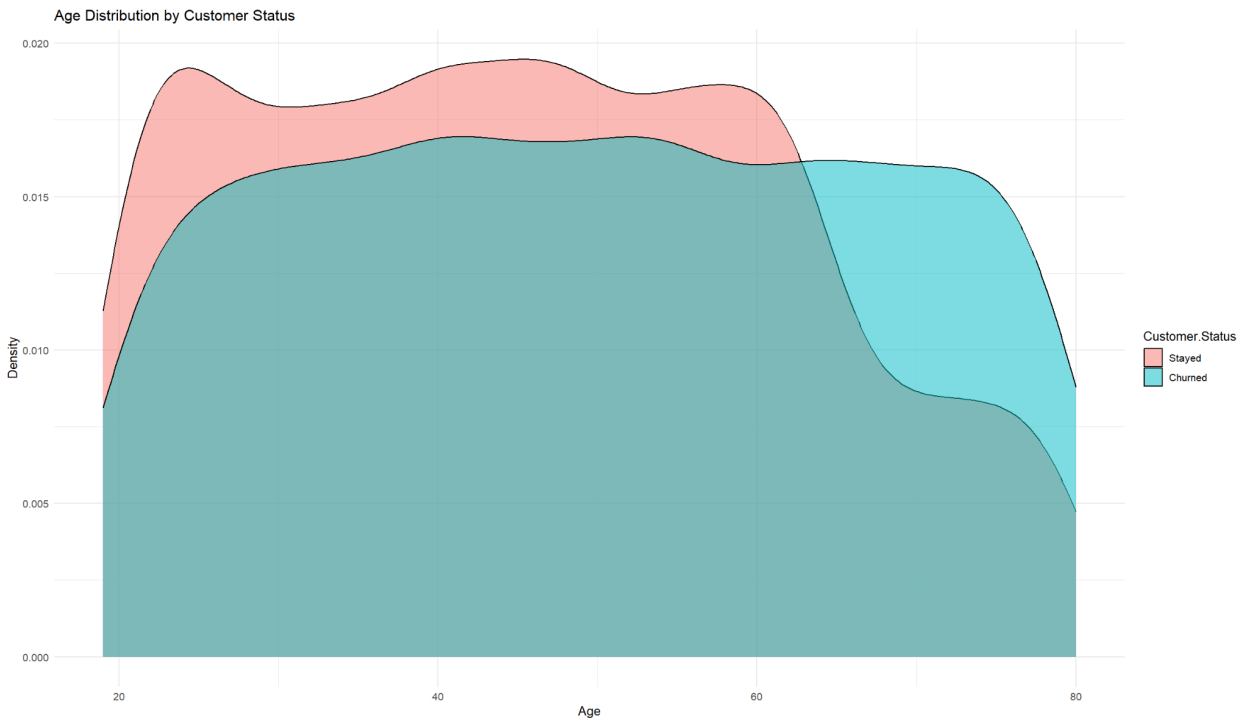
- Median Tenure: The line inside the green box is barely visible above zero, showing that the median tenure of joined customers is very low.
- IQR: The green box is almost flat, indicating that all joined customers have almost the same tenure, close to 0 months.

(**The factor *Joined*; has been removed in the new version of my R code since it was not the factor we were trying to predict.**)

Stayed Customers:

- Median Tenure: The line within the blue box shows that the median tenure of stayed customers is approximately 40 months.
- IQR: The blue box has a tenor of approximately 25 to 60 months, showing the broad tenure of stayed customers.

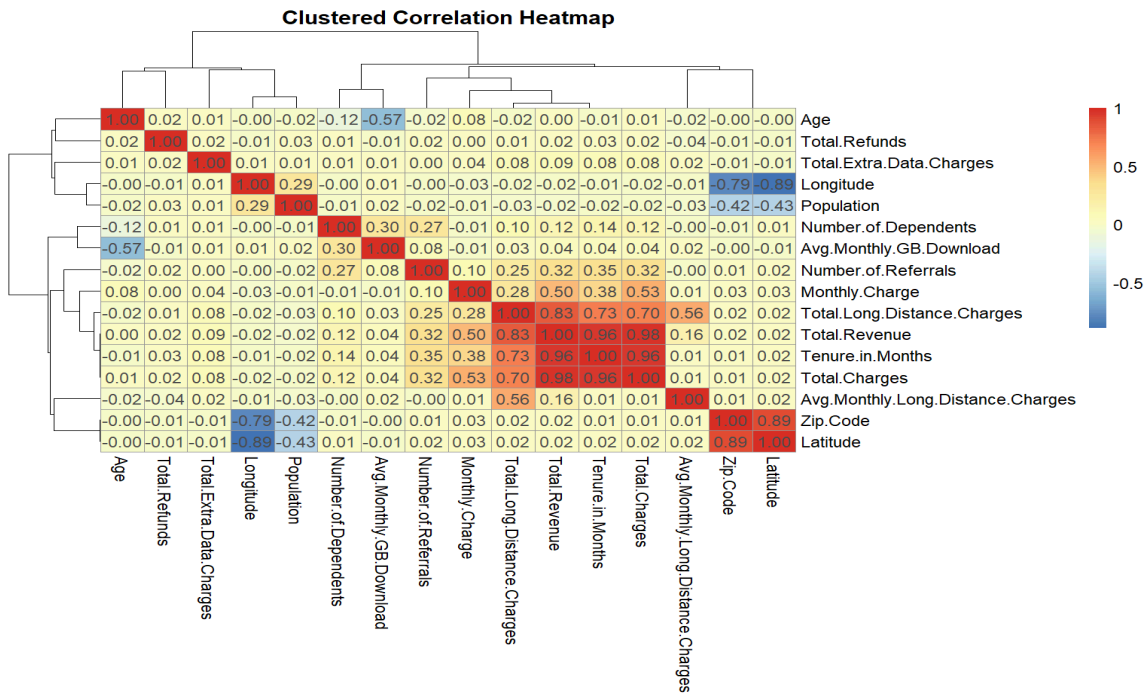
4.5 Density Plot for age distribution by customer status:



⬆️ As can be seen from the chart above, there is a fairly even distribution of retained and churn customers across most age ranges (20 to 60). The density of churn customers is slightly higher for younger age groups. For customers over 60, the density of those who churn gradually decreases compared to those who stay.

Hence, younger customers are more likely to churn than older customers, and churn rates decrease with age, especially for customers over 60. This suggests that younger customers may be more likely to leave the service, while older customers are more likely to stay. Therefore, telecom companies can strengthen their sales strategies for young people and make changes.

4.6 Heatmap with hierarchical clustering:

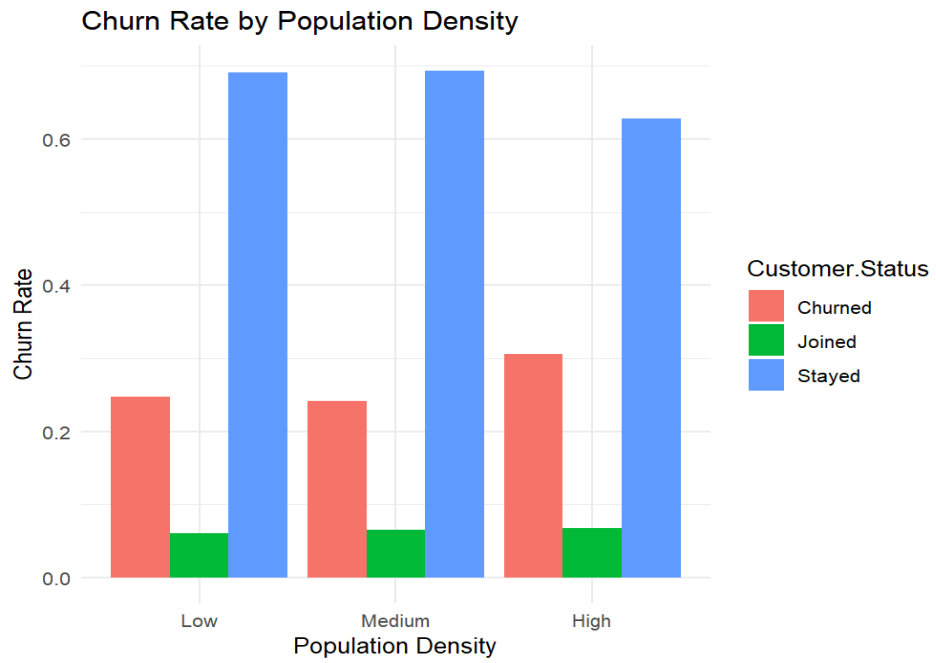


⬆️ As can be seen from the chart above, some variables like Total. Charges, Total. Revenue, and Tenure.in.Months are very strongly correlated, suggesting they share a strong linear relationship.

Total. Long. Distance. Charges also show high correlation with Total. Revenue and Total Charges, which is logical, as higher usage typically comes with higher fees, thus increasing overall revenue.

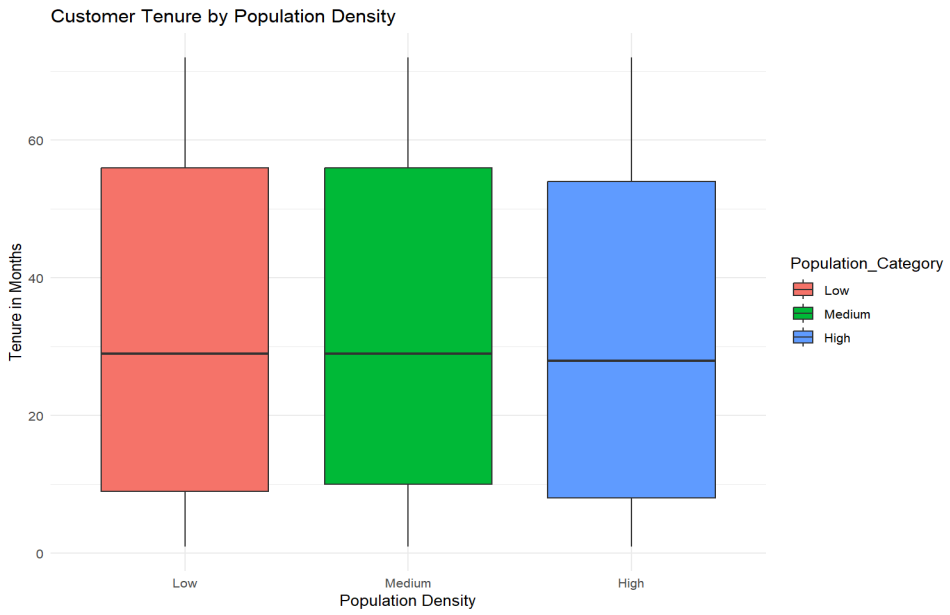
Variables such as Population and Latitude show a strong negative correlation with Zip. Code. This shows that geographical and demographic patterns might influence the distribution of zip codes and population densities.

4.7 Churn rate by population density:



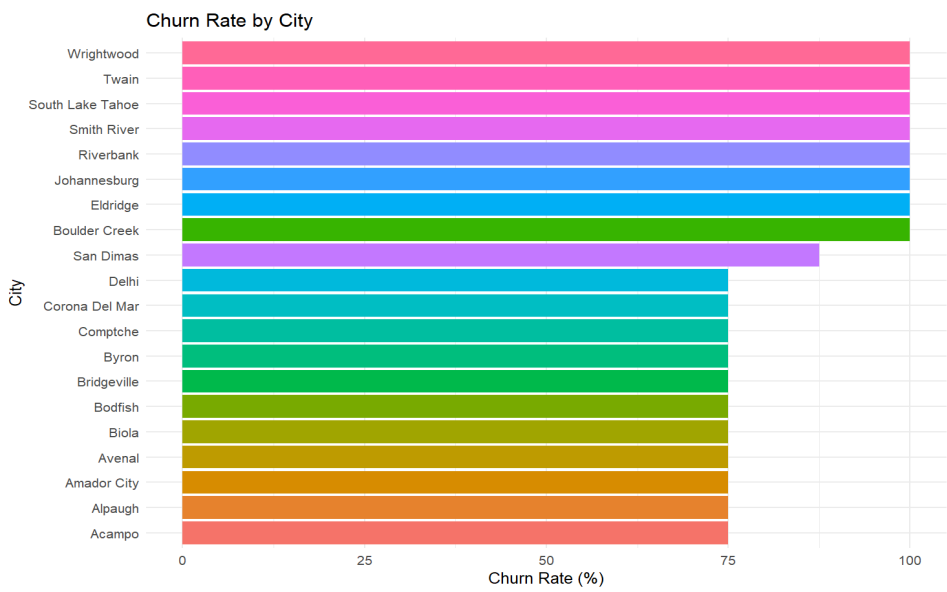
⬆️ As can be seen from the chart above, according to the classification of population data (0, 0.33, 0.66, 1), population density is divided into three levels: low, medium, and high. The distribution of customers in low and medium areas is similar. In areas with high population density, the churned customer increased, indicating that higher population density may be associated with higher churn rates and need to strengthen retention efforts. (The factor *Joined*; has been removed in the new version of my R code since it was not the factor we were trying to predict)

4.8 Box plot of tenure by population density:



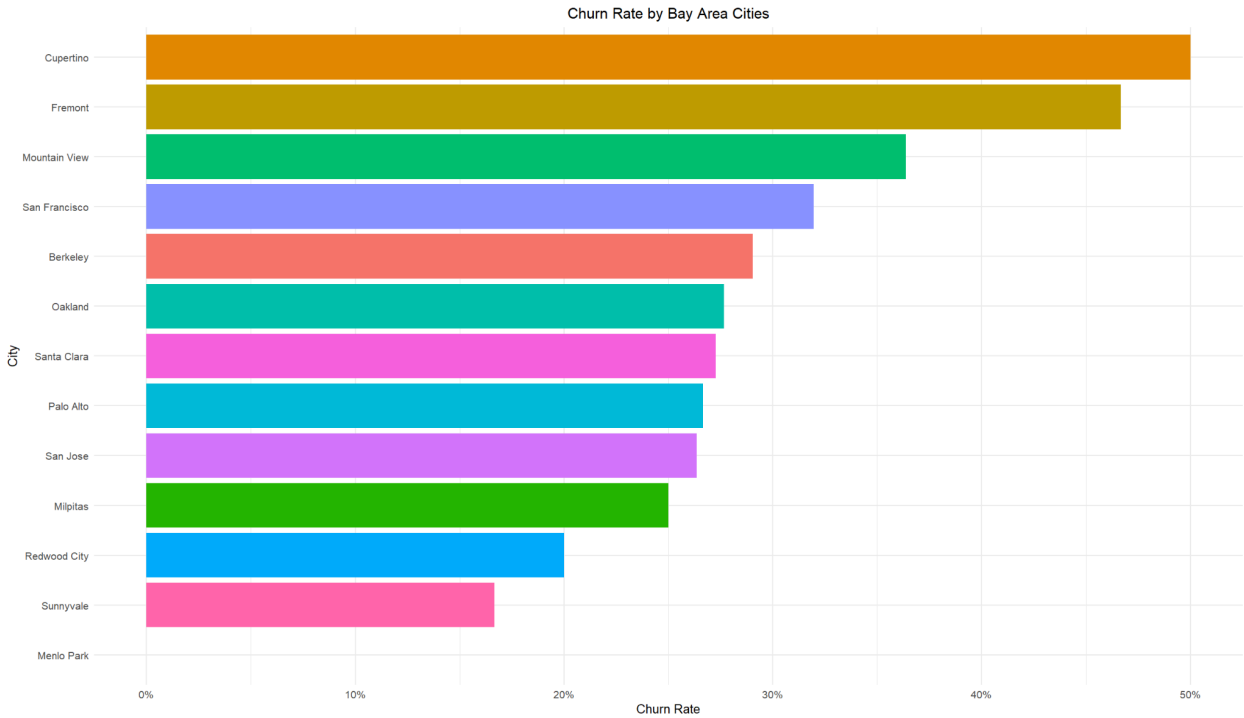
⬆️ As can be seen from the chart above, this box plot shows no significant differences in areas of different densities but is slightly lower in high-density areas. This may simply indicate that, on average, customers in higher population density areas stay slightly shorter. (**This chart has been removed in the new version of my R code since it was not necessary**)

4.9 Churn rate by top 20 cities:



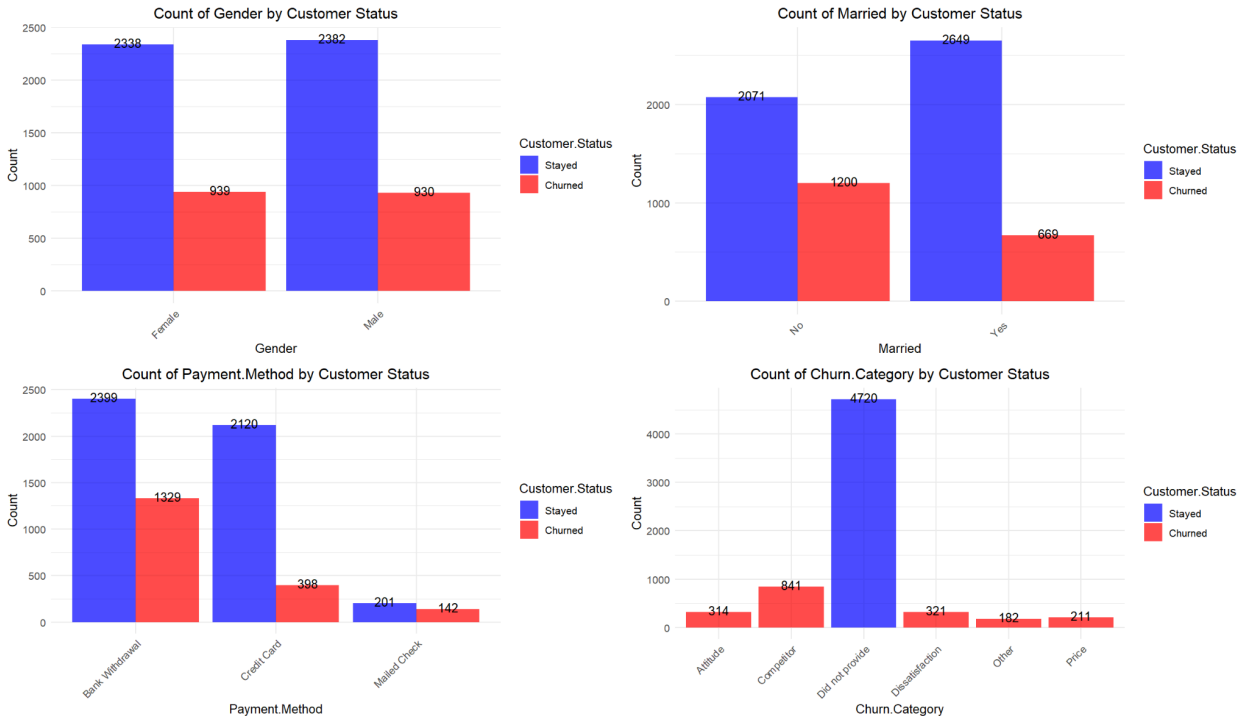
⬆️ As can be seen from the chart above, cities such as Boulder Creek, Eldridge, Johannesburg, Riverbank, Smith River, South Lake Tahoe, Twain, and Wrightwood all had 100% churn rates, causing by too small sample sizes. But this chart still represents an interesting insight. In particular, the changing of geographic or demographic factors, or probably because of local competition and service quality, affect customer retention. However, even because there is less data in these areas, telecom companies can still develop some effective strategies to activate potential customers.

4.10 Churn rate by bay area cities:



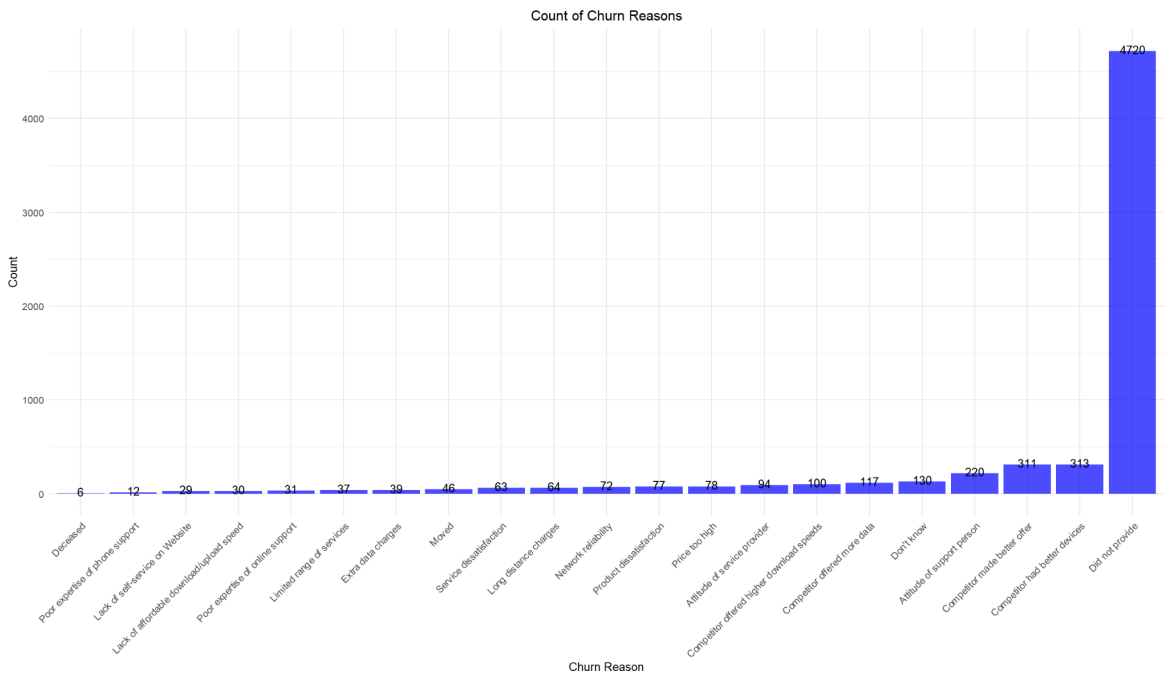
⬆️ As can be seen from the chart above, we can see the results for the top bay Area cities. Due to insufficient or too small population data in some cities, the churn rate may be high. But it's still an indicator worth considering. Perhaps for areas with high churn rates, potential actions we would recommend to telecom companies may include improving customer service, offering better offers or addressing specific issues faced by customers in these areas to find new customers.

4.11 Categorical features by customer status:



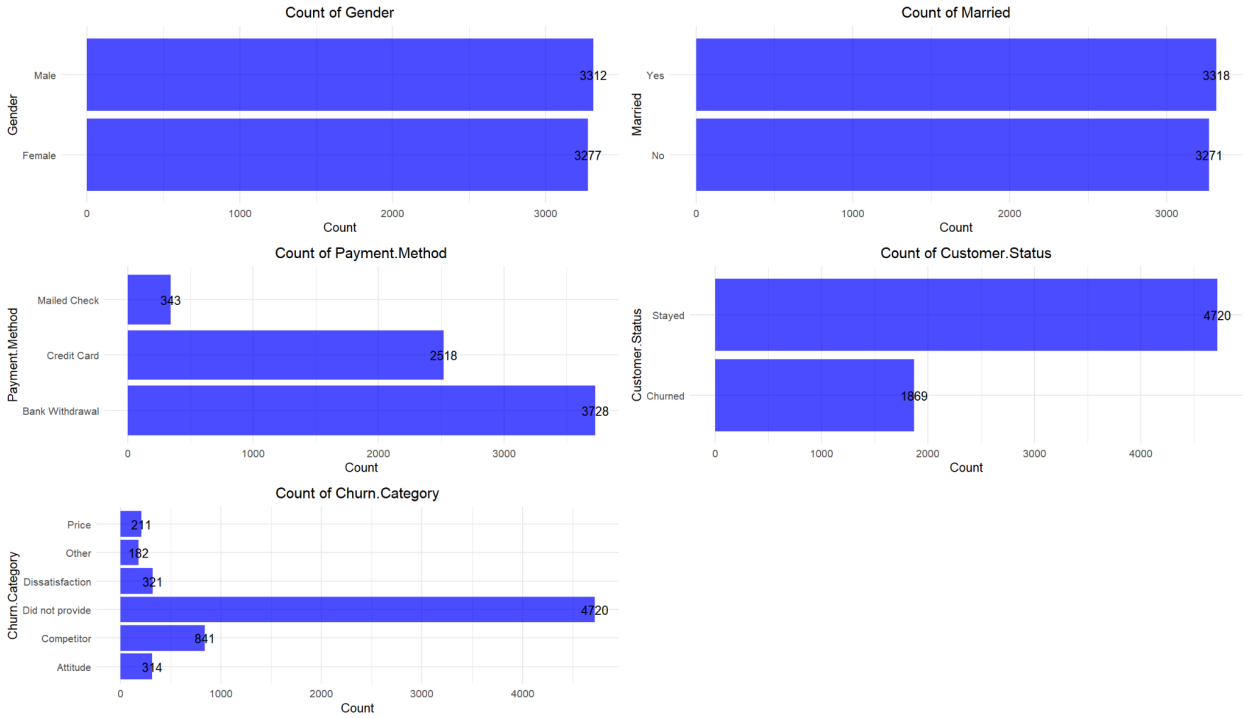
⬆️ As can be seen from the chart above, gender does not significantly affect churn rates, but marital status does, with unmarried customers having higher churn rates. And customers who use banks to withdraw money are more likely to churn than customers who use credit cards. Finally, while many customers did not provide a reason for churn, but for those who did, competition and dissatisfaction were significant factors.

4.12 Count of churn reasons:



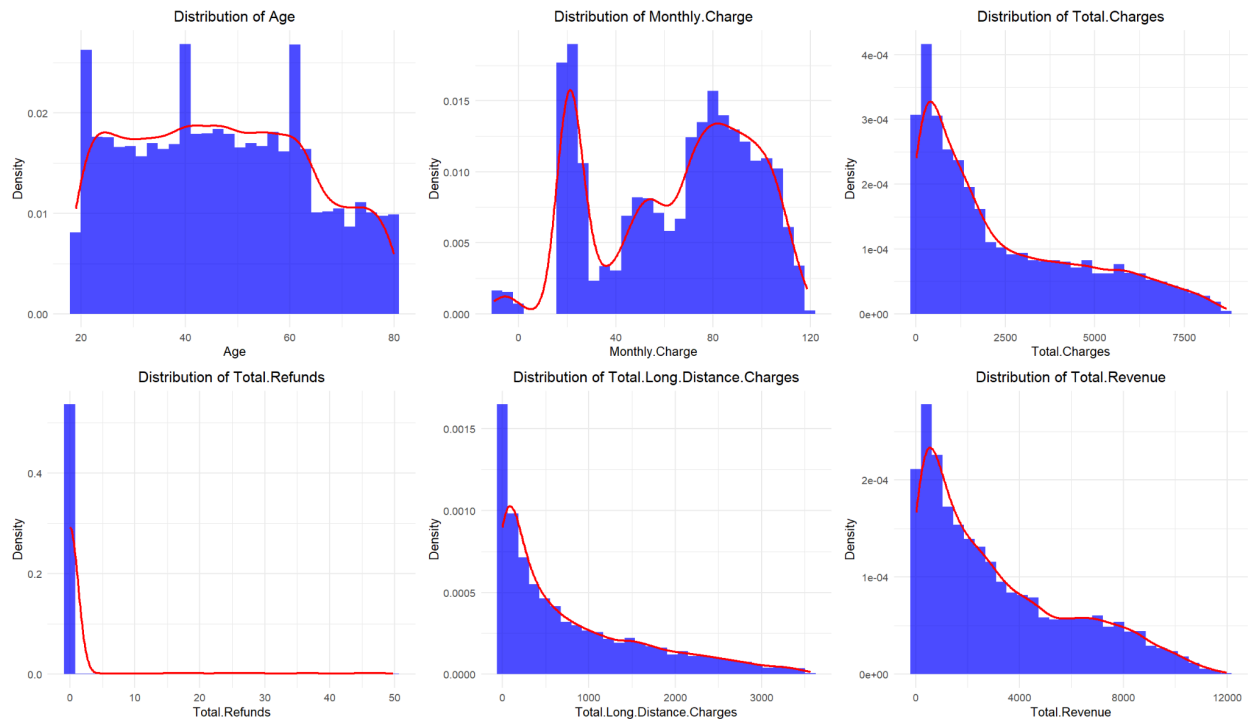
⬆️ As can be seen from the chart above, most customers did not provide any reason for churn. This lack of information makes it difficult to directly address specific issues. However, **WenZhou** proposed an interesting method in class to make this null value replenished, thanks for his helpful advice. The other two important influencing factors are the attitude of competitors and customer service personnel, which means that telecom companies can better provide competitive products and prices, and they can also do more training for the people in charge of customer service.

4.13 Distribution of categorical features:



⬆️ As can be seen from the chart above, similar to the results of chart 4.11, the distribution of gender and marital status is very balanced. Bank withdrawal is the preferred payment method and has the highest customer churn rate. Customer status is the variable we want to predict, and out of our 7043 observations, there are 1869 customer churned. From the reasons for churn, a large proportion of customers did not provide a reason for churn, but reasons related to competitors and dissatisfaction with service attitude were the most common as the same insight from chart 4.11.

4.14 Distribution of numerical features:



⬆️ As can be seen from the chart above, the customers are evenly distributed across different age groups, but has three peaks in certain age groups. Monthly charge has two main customer segments, showing different service plans or pricing strategies. The other three variables are highly right-skewed, which means that most of the revenue and charges are generated by only a few customers who are large consumer groups.

4.15 Key insights from the EDA:

Based on the above EDA and combined with the first two question (Churn Finding & Customer Segmentation), we can initially gain the following insights:

- Tenure: Customers with longer tenure are more likely to stay.
- Monthly Charges: The distribution of monthly charges suggests that customers paying higher monthly fees may have varied service plans. And most customers who use more additional service would likely stay.

- Churn Reasons: Customers leaving due to competitor offers, dissatisfaction, and high prices highlight the importance of competitive pricing and service quality.
- Payment Methods: Customers using credit cards have lower churn rates compared to those using bank withdrawals.
- Service Usage: Higher long-distance charges and total charges are correlated with higher revenue and longer tenure.

➢ **Recommendations for Retention:**

Based on the findings, we recommend that telecom companies implement programs that reward long-term customers to extend their tenure, as well as encourage credit card payments. Next, focus on addressing the reasons for high churn rates in different groups, and use enhanced service offerings and personalized customer service to improve satisfaction. Additionally, provide customized offers and services to young or single customers to reduce their churn rate. Finally, target high-churn areas in specific cities like Cupertino, Fremont, and Mountain View. Telecom companies can also leverage location-based segmentation and targeted marketing efforts to effectively reach new potential customers through localized advertising.

5. Logistic Regression (by Chandana)

When attempting to forecast the likelihood of a binary result, a statistical technique called logistic regression is frequently used: binary classification problems. Logistic regression is designed for scenarios in which the dependent variable has only two possible values, in contrast to linear regression, which works with continuous outcomes. Logistic regression maps input variables to a probability score between 0 and 1, as opposed to fitting a straight line to the data. This function is sometimes referred to as the sigmoid function. This function makes sure that the

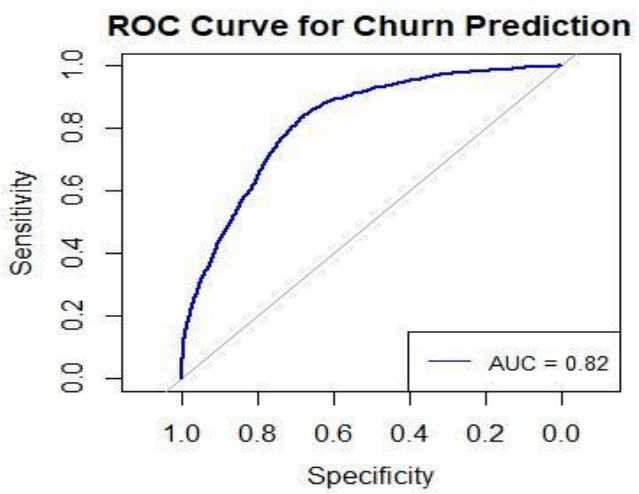
expected probabilities fall within the acceptable probability range. Logistic regression is used in model training to obtain coefficients that show how input factors affect the result.

5.1 ROC (Receiver Operating Characteristic)

A graph called the ROC curve illustrates how well a model can differentiate between two classes—in your instance, churning and non-churning clients. The following is what the axes stand for:

The Specificity X-axis indicates how well the model identifies non-churning clients, or those who won't leave. High specificity is shown by the model's exceptional performance in this area.

Sensitivity, or the Y-axis, displays how well the model predicts which consumers will leave (churning customers). The model has high sensitivity if it performs this well.



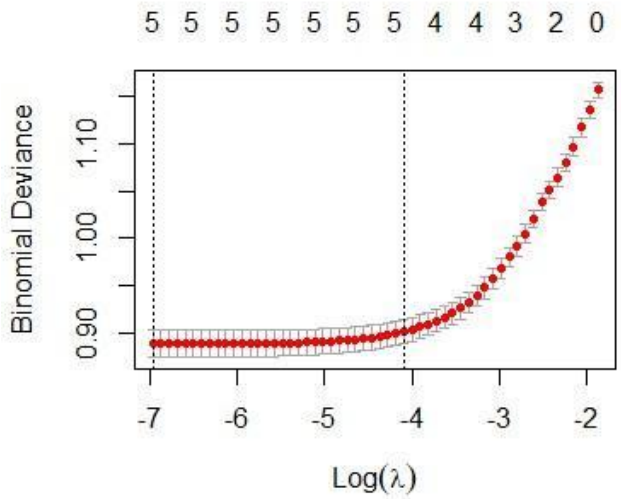
When the ROC curve gracefully traces the upper left corner of the graph, it signifies the model's exceptional performance in two critical aspects: firstly, it's highly sensitive in identifying disgruntled clients, ensuring that a significant proportion of churning customers are correctly recognized. Secondly, it demonstrates high specificity by accurately determining which clients are not churning, effectively reducing false alarms.

An AUC (Area Under the Curve) value of 0.82 underscores this excellence, indicating an 82% likelihood that the model will accurately discern customers likely to churn from those who won't. Generally, as the AUC approaches 1, the model's discriminatory power improves, suggesting better performance overall.

In summary, the model excels in differentiating between customers at risk of leaving and those likely to stay, showcasing both high sensitivity and specificity.

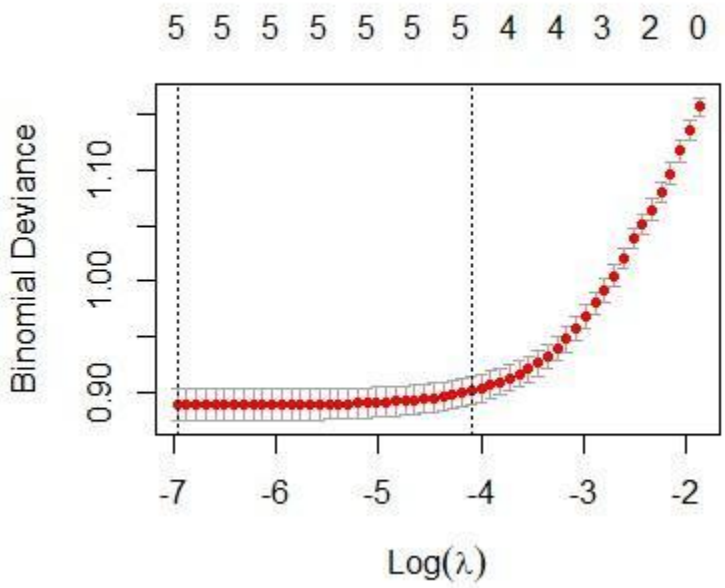
6. Lasso Regression

Lasso Regression, short for Least Absolute Shrinkage and Selection Operator, augments the linear regression equation with a penalty term proportional to the absolute magnitude of the coefficients multiplied by a constant (alpha). This addition facilitates feature selection by encouraging less significant coefficients to shrink to zero. Consequently, Lasso Regression effectively prunes out less important features, streamlining the model and enhancing interpretability. Particularly useful in datasets rife with numerous features, Lasso Regression automates the identification of the most relevant predictors while disregarding irrelevant ones, thereby improving model efficiency and performance.



7. Ridge Regression

In a manner akin to Lasso, Ridge Regression enhances the linear regression equation by incorporating a penalty term. However, unlike Lasso, the penalty term in Ridge Regression is proportional to the squared magnitude of the coefficients multiplied by a constant (alpha). This distinction allows Ridge Regression to penalize large coefficients without necessitating their precise reduction to zero, as Lasso does. Instead, Ridge Regression induces coefficients to gradually diminish toward zero while retaining them within the model. This property makes Ridge Regression particularly effective in mitigating multicollinearity, where predictor variables exhibit high correlation. By tempering the variance in the model's predictions, Ridge Regression fosters model stability and serves as a bulwark against overfitting, ultimately enhancing the model's robustness and generalization performance.



8. Conclusion

The above is a preliminary analysis of the telecom churned customers for the final project as of Module 4. We reviewed reports from several other authors on churn prediction and customer segmentation and implemented them in our dataset. In this analysis, important factors for churn include tenure, contract type and so on, with longer tenure and fixed-term contracts significantly reducing churn risk. Companies can offer favorable promotions to customers with short-term contracts and encourage them to become fixed-term contract customers.

Additionally, customer segmentation analysis reveals different groups based on demographics and spending, which helps in developing targeted marketing strategies. However, high churn rates in certain cities and specific demographic groups require more tailored retention plans and strategies with proactive customer service. This is important to allow telecom companies to segment the market and improve customer retention with high satisfaction.

The logistic regression analysis uses ROC (Receiver Operating Characteristic) and AUC (Area Under the Curve) to evaluate the model.

The x-axis represents the True Negative Rate (TNR), indicating the proportion of non-churning customers (negative class). The y-axis represents the True Positive Rate (TPR), indicating the proportion of churning customers (positive class). The ROC curve "hugs the upper left corner of the graph," indicating a model with 100% TPR and 100% TNR.

The AUC of 0.82 suggests that the model is good at distinguishing between customers who will churn and those who won’t. The logistic regression analysis using ROC and AUC indicates that the model is effective in predicting customer churn, with an AUC of 0.82 showing good performance in distinguishing between churning and non-churning customers.

9. References

1. Telecom Customer Churn Prediction Dataset:

   https://www.kaggle.com/datasets/shilongzhuang/telecom-customer-churn-by-maven-analytics

2. Predict Customer Churn with R:

   https://towardsdatascience.com/predict-customer-churn-with-r-9e62357d47b4

3. Tutorial: Use R to create, evaluate, and score a churn prediction model:

   https://learn.microsoft.com/en-us/fabric/data-science/r-customer-churn

4. Telco Customer Churn Prediction in R:

   https://www.kaggle.com/code/farrahxym/telco-customer-churn-prediction-in-r

5. Telco Customer Churn Rate Analysis:

   https://towardsdatascience.com/telco-customer-churnrate-analysis-d412f208cbbf

6. Telco Customer Churn Prediction Using Machine Learning and Deep Learning:

   https://medium.com/@zulfikarirham02/telco-customer-churn-prediction-using-machine-learning-and-deep-learning-8d1905b04980

10. R Code

10.1 EDA (by Cheng)

```
#-----------------#

# ALY6015 - Intermediate Analytics @Silicon Valley Campus #

# Final Project - Telecom Customer Churn Analysis #

# May.05.2024 #

# Cheng Liu #

#-----------------#

# Import the library

library(tibble)

library(ggplot2)

library(tidyverse)

library(gridExtra)

library(pheatmap)

library(dplyr)
```

```
# Load the datasets

telecom <- read.csv("Telecom Customer Churn Prediction
Dataset/telecom_customer_churn.csv")

zipcode <- read.csv("Telecom Customer Churn Prediction
Dataset/telecom_zipcode_population.csv")

# Merge the datasets based on Zip Code

merged_telecom <- merge(telecom, zipcode, by.x = "Zip.Code", by.y = "Zip.Code", all.x =
TRUE)

merged_telecom <- as_tibble(merged_telecom)

# Check the datasets

str(merged_telecom)

summary(merged_telecom)

# Check for missing values

sapply(merged_telecom, function(x) sum(is.na(x)))

# Remove joined for customer status

merged_telecom <- merged_telecom %>% filter(Customer.Status != 'Joined')

# Numerical features

numerical_features <- c('Age', 'Monthly.Charge','Total.Charges', 'Total.Refunds',

                        'Total.Long.Distance.Charges', 'Total.Revenue')

# Plot distribution of numerical features
```

```
numerical_plots <- lapply(numerical_features, function(feature) {
  ggplot(merged_telecom, aes_string(x = feature)) +
    geom_histogram(aes(y = ..density..), bins = 30, fill = "blue", alpha = 0.7) +
    geom_density(col = "red", size = 1) +
    ggtitle(paste("Distribution of", feature)) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.5)) +
    labs(x = feature, y = "Density")
})
grid.arrange(grobs = numerical_plots, ncol = 3)
# Categorical features
merged_telecom$Churn.Category[merged_telecom$Churn.Category == ""] <- "Did not provide"
categorical_features <- c('Gender', 'Married', 'Payment.Method', 'Customer.Status',
'Churn.Category')
# Plotting distribution of categorical features
categorical_plots <- lapply(categorical_features, function(feature) {
  ggplot(merged_telecom, aes_string(x = feature)) +
    geom_bar(fill = "blue", alpha = 0.7) +
    ggtitle(paste("Count of", feature)) +
```

```
    theme_minimal() +
    geom_text(stat = 'count', aes(label = ..count..), vjust = 0.5) +
    theme(plot.title = element_text(hjust = 0.5)) +
    labs(x = feature, y = "Count") +
    coord_flip()
})
grid.arrange(grobs = categorical_plots, ncol = 2)
# Bar plot for categorical features by customer status
categorical_features <- c('Gender', 'Married', 'Payment.Method', 'Churn.Category')
merged_telecom$Customer.Status <- factor(merged_telecom$Customer.Status, levels =
c("Stayed", "Churned"))
# Create individual plots for each categorical feature
categorical_plots <- lapply(categorical_features, function(feature) {
  ggplot(merged_telecom, aes_string(x = feature, fill = "Customer.Status")) +
    geom_bar(position = 'dodge', alpha = 0.7) +
    geom_text(stat = 'count', aes(label = ..count..), position = position_dodge(width = 0.9), vjust =
0.3) +
    ggtitle(paste("Count of", feature, "by Customer Status")) +
    theme_minimal() +
```

```
    theme(plot.title = element_text(hjust = 0.5), axis.text.x = element_text(angle = 45, hjust = 1))
+
    labs(x = feature, y = 'Count') +
    scale_fill_manual(values = c("blue", "red"))
})
grid.arrange(grobs = categorical_plots, ncol = 2)
# Calculate counts of churn reason and reorder factor levels by count
merged_telecom$Churn.Reason[merged_telecom$Churn.Reason == ""] <- "Did not provide"
churn_reason <- merged_telecom %>%
  count(Churn.Reason) %>%
  arrange(n)
# Convert churn reason to a factor with levels ordered by count
merged_telecom$Churn.Reason <- factor(merged_telecom$Churn.Reason, levels =
churn_reason$Churn.Reason)
# Bar plot for count of churn reason
ggplot(merged_telecom, aes(x = Churn.Reason)) +
  geom_bar(fill = "blue", alpha = 0.7) +
  geom_text(stat = 'count', aes(label = ..count..), vjust = 0.3) +
  ggtitle("Count of Churn Reasons") +
```

```
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5), axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = 'Churn Reason', y = 'Count')
# Bar plot for contract vs. customer status
ggplot(merged_telecom, aes(x = Contract, fill = Contract)) + geom_bar() + facet_wrap(~
Customer.Status)
# Box plot for tenure in months vs. customer status
ggplot(merged_telecom, aes(x = Customer.Status, y = Tenure.in.Months, fill = Customer.Status))
+
  geom_boxplot() +
  labs(title = "Tenure in Months vs. Customer Status", x = "Customer Status", y = "Tenure in
Months") +
  theme_minimal()
# Density Plot for age distribution by customer status
ggplot(merged_telecom, aes(x = Age, fill = Customer.Status)) +
  geom_density(alpha = 0.5) +
  labs(title = "Age Distribution by Customer Status", x = "Age", y = "Density") +
  theme_minimal()
# Calculate the correlation matrix
numeric_vars <- merged_telecom[, sapply(merged_telecom, is.numeric)]
```

```
correlation_matrix <- cor(numeric_vars, use = "complete.obs")

correlation_matrix

# Plot the heat map with hierarchical clustering

pheatmap(correlation_matrix, clustering_distance_rows = "correlation", clustering_distance_cols
= "correlation",

        display_numbers = TRUE, fontsize_number = 10, main = "Clustered Correlation
Heatmap")

# Churn rate by population density

# Define low & medium & high population category

population_categories <- cut(merged_telecom$Population, breaks =
quantile(merged_telecom$Population, probs = c(0, 0.33, 0.66, 1)),

                        labels = c("Low", "Medium", "High"), include.lowest = TRUE)

merged_telecom <- merged_telecom %>%

  mutate(Population_Category = population_categories)

# Calculate churn rate by population category

churn_rate_by_pop <- merged_telecom %>%

  group_by(Population_Category, Customer.Status) %>%

  summarise(Count = n()) %>%

  mutate(Churn_Rate = Count / sum(Count))

# Plot churn rate by population category
```

```
ggplot(churn_rate_by_pop, aes(x = Population_Category, y = Churn_Rate, fill =
Customer.Status)) +
  geom_bar(stat = "identity", position = position_dodge()) +
  labs(title = "Churn Rate by Population Density", x = "Population Density", y = "Churn Rate") +
  theme_minimal()
# Calculate churn rate by city
churn_rate_by_city <- merged_telecom %>%
  group_by(City) %>%
  summarise(Total_Customers = n(),
            Churned_Customers = sum(Customer.Status == "Churned"),
            Churn_Rate = Churned_Customers / Total_Customers * 100) %>%
  arrange(desc(Churn_Rate))
# Plot churn rate by top 20 city
ggplot(churn_rate_by_city[1:20, ], aes(x = reorder(City, Churn_Rate), y = Churn_Rate, fill =
City)) +
  geom_bar(stat = "identity") +
  labs(title = "Churn Rate by City", x = "City", y = "Churn Rate (%)") +
  coord_flip() +
  theme_minimal() +
```

```
  theme(legend.position = "none")

# Define bay area cities

bay_area_cities <- c("San Francisco", "Oakland", "San Jose", "Berkeley", "Fremont",

                     "Palo Alto", "Sunnyvale", "Santa Clara", "Mountain View",

                     "Redwood City", "Menlo Park", "Cupertino", "Milpitas")

# Filter the dataset for bay area cities

merged_telecom_bay_area <- merged_telecom %>% filter(City %in% bay_area_cities)

# Calculate churn rate by city within Bay Area

churn_rate_by_bay_area <- merged_telecom_bay_area %>%

  group_by(City) %>%

  summarize(

    Total_Customers = n(),

    Churned_Customers = sum(Customer.Status == "Churned"),

    Churn_Rate = Churned_Customers / Total_Customers

  )

# Plot churn rate by bay area city

ggplot(churn_rate_by_bay_area, aes(x = reorder(City, Churn_Rate), y = Churn_Rate, fill =
City)) +

  geom_bar(stat = "identity") +
```

```
  coord_flip() +
  ggtitle("Churn Rate by Bay Area Cities") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5)) +
  labs(x = 'City', y = 'Churn Rate') +
  theme(legend.position = "none") +
  scale_y_continuous(labels = scales::percent)
# Plot customer status by Bay Area city
ggplot(merged_telecom_bay_area, aes(x = City, fill = Customer.Status)) +
  geom_bar(position = "dodge", alpha = 0.7) +
  coord_flip() +
  ggtitle("Customer Status by Bay Area Cities") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5)) +
  labs(x = 'City', y = 'Count') +
  scale_fill_manual(values = c("blue", "red"))


# Stacked bar plot customer status by Bay Area city
ggplot(merged_telecom_bay_area, aes(x = City, fill = Customer.Status)) +
```

```
  geom_bar(position = "fill", alpha = 0.7) +

  coord_flip() +

  ggtitle("Customer Status by Bay Area Cities") +

  theme_minimal() +

  theme(plot.title = element_text(hjust = 0.5)) +

  labs(x = 'City', y = 'Proportion') +

  scale_fill_manual(values = c("blue", "red")) +

  scale_y_continuous(labels = scales::percent)
```

10.2 Logistic Regression (by Chandana)

```
# Import the library

library(dplyr)

# Load the datasets

telecom_customer_churn <- read_csv("final project/week 6/telecom_customer_churn.csv")

telecom_zipcode_population <- read_csv("final project/week

6/telecom_zipcode_population.csv")

# Merge the datasets based on Zip Code

merged_telecom <- left_join(telecom_customer_churn, telecom_zipcode_population, by = "Zip

Code")

# Check if the merge was successful

head(merged_telecom)

# Check column names of merged_telecom

colnames(merged_telecom)
```

```
# Check column names of merged_telecom
colnames(merged_telecom)
# Check unique values of Customer Status
unique(merged_telecom$`Customer Status`)
# Recode Customer Status into binary outcome variable (Churned = 1, Not churned = 0)
merged_telecom$Churn <- ifelse(merged_telecom$`Customer Status` == "Churned", 1, 0)
# Perform logistic regression with the new binary outcome variable (Churn)
logistic_model <- glm(Churn ~ Contract + `Tenure in Months` + Age + Population,
                      data = merged_telecom, family = binomial)
# Summarize the logistic regression model
summary(logistic_model)
# Install and load the pROC package (if not already installed)
install.packages("pROC")
library(pROC)
# Load the datasets
telecom_customer_churn <- read_csv("final project/week 6/telecom_customer_churn.csv")
telecom_zipcode_population <- read_csv("final project/week
6/telecom_zipcode_population.csv")
# Predict probabilities of churn using the logistic regression model
predictions <- predict(logistic_model, type = "response")
# Create a ROC curve
roc_obj <- roc(merged_telecom$Churn, predictions)
# Plot ROC curve
plot(roc_obj, main = "ROC Curve for Churn Prediction", col = "blue", lwd = 2)
```

```
# Add AUC (Area Under the Curve) to the plot

auc_val <- round(auc(roc_obj), 2)

legend("bottomright", legend = paste("AUC =", auc_val), col = "blue", lty = 1, cex = 0.8)

# Install and load the glmnet package (if not already installed)

install.packages("glmnet")

library(glmnet)

# Create the outcome vector y

y <- ifelse(merged_telecom$`Customer Status` == "Churned", 1, 0)

# Create the predictor matrix X

X <- model.matrix(~ Contract + `Tenure in Months` + Age + Population, data =

merged_telecom)

# Perform LASSO logistic regression

lasso_model <- cv.glmnet(X, y, family = "binomial", alpha = 1)=

# Plot the cross-validated mean squared error (mse) vs. log(lambda)

plot(lasso_model)

# Perform ridge logistic regression

ridge_model <- cv.glmnet(X, y, family = "binomial", alpha = 0)

# Plot the cross-validated mean squared error (mse) vs. log(lambda)

plot(ridge_model)
```